Workshop Report

Meningitis Metadata Standard — Standards Stakeholders Meeting

04–07 May 2026

PHA4GE Data Standards Initiative

1. Background and Purpose

This report documents the proceedings of a four-day standards stakeholders workshop convened to advance the development of the Meningitis Metadata Standard. The workshop was the second stakeholder meeting for this standard, the first having taken place in October 2025, and represented the first extended working session to develop the data dictionary in detail.

The meningitis standard is one of several pathogen-specific standards being developed under the PHA4GE data standardisation initiative. At the time of this workshop, the meningitis standard was at an earlier stage of development than the other standards in this series, and the primary objective was therefore to make substantive progress on the metadata data dictionary, controlled vocabularies, and field tiering across all four domains.

The four domains the workshop was structured around were:

- Case/Patient
- Specimen
- Laboratory/Diagnostics, and
- Bioinformatics.

Each day built on the previous, progressing from field-by-field dictionary finalisation through to the population of controlled vocabulary picklists and the assignment of bioinformatics serotyping and AMR gene lists.

Note: This report covers the meningitis workshop only, and will form one chapter of the consolidated workshop report that also covers the malaria, neonatal sepsis, and cholera workshops.

2. Participants

Name	Affiliation	Country
Stanford Kwenda (Lead data standard developer + Facilitator)	PHA4GE	South Africa
Buhle Ntozini (co-facilitator)	SANBI, University of the Western Cape	South Africa
Fatima Ly	Institut Pasteur de Dakar	Senegal
Moussa Diagne	Institut Pasteur de Dakar	Senegal
Chifundo Salifu	Malawi Liverpool Wellcome Research Programme	Malawi
Susan Meiring	National Institute for Communicable Diseases (NICD)	South Africa
Rudzani Mashau	Wits Mycology, a Division of Wits Health Consortium	South Africa
Yiyé André Nagalo	Université Lédéa Bernard Ouédraogo de Ouahigouya	Burkina Faso
Molebogeng Kolojane	National Health Laboratory Services (NHLS)	South Africa

Note takers: Rekha Sathyan (lead), Harrison Abrahams, and Justice Trésor Ngom.

3. Workshop Overview

The workshop addressed a metadata standard covering bacterial, viral, and fungal pathogens causing meningitis, with a focus on community-acquired disease. The standard is structured around four domains:

Table 1: *Overview of the four metadata domains addressed during the workshop.*

Domain	Description
Case / Patient	Patient demographics, clinical presentation, vaccination status, outcomes, and travel history
Specimen	Specimen type, collection device, transport, and storage
Laboratory / Diagnostics	Macroscopic and microscopic results, biochemistry, culture, AST, serotyping, molecular testing, and sequencing

Domain	Description
Bioinformatics	Assembly, QC metrics, MLST, serotyping, virulence genes, and AMR gene detection

A tiering system was applied throughout: mandatory fields are required for minimum data usability; recommended fields are strongly advised for best results; and optional fields are voluntary and context-dependent. The overarching aim is to minimise free text and maximise the use of controlled vocabularies to enable machine-actionable, interoperable data.

The colour coding used in the Tiering column of tables in this report is as follows:

Mandatory	Recommended	Optional	Free text summary

4. Day 1 (04 May 2026)

4.1 Opening Discussion: Purpose, Scope, and Governance

The workshop opened with an introduction by Stanford Kwenda, followed by a presentation on why metadata standards are needed for meningitis surveillance. Key themes included the interpretation gap between genomic, molecular, and diagnostic data; the challenge of heterogeneous data across settings; and the importance of controlled vocabularies and ontology alignment for machine-actionable data sharing.

An open discussion followed, covering the goals of the standard, implementation challenges, and the relationship between the meningitis and neonatal sepsis standards. Key outcomes from this discussion are summarised below.

Purpose and governance

The standard is intended to harmonise data reporting and support data sharing and governance between laboratories and countries during epidemic outbreak response. While participants recognized the value of a common standard, concerns were raised regarding adoption, including implementation challenges, legacy systems, and the risk of over-engineering.

Several practical implementation pathways were identified to facilitate uptake, including spreadsheet-based solutions (e.g., DataHarmonizer), standardized data exchange formats, and direct integration into existing systems and workflows.

Although there is no formal mandate for governments to adopt the standard, adoption will be encouraged through advocacy and supporting resources. Africa CDC may further facilitate uptake through integration of the standard into AGARI. Additional dissemination activities, including a manuscript and supporting documentation, will be developed to promote awareness and adoption within the public health community.

Alignment with the Neonatal Sepsis Standard

An early discussion focused on the overlap between the meningitis standard and the neonatal sepsis guidance currently being developed by PHA4GE. Participants noted that several data elements are common across both standards, including sample identifiers, geolocation, and collection dates, and questioned whether a single merged standard could reduce redundancy.

It was agreed that the standards will remain disease-specific to meet distinct surveillance and public health needs. However, alignment will be pursued where appropriate to minimize duplication, improve consistency, and increase implementation efficiency. A number of shared fields were identified and will be harmonized across both standards where possible.

Scope of Pathogens

Participants sought clarification on the scope of the standard and specifically asked why tuberculosis meningitis was not included. It was noted that the pathogen list had been agreed during a previous meeting (October 2025) and remains a work in progress. The current scope focuses on community-acquired meningitis and includes bacterial, viral, and fungal pathogens.

Workshop Outputs

The workshop outputs were confirmed as:

- A finalized data dictionary;
- Controlled vocabularies for selected fields;
- Ontology mappings, where feasible; and
- A pilot-ready standard, where possible.

4.2 Dictionary Finalisation — Case / Patient Domain

Following the opening discussion on Day 1, the group began working through the Case/Patient domain line by line. In this report, the Tiering column is populated only where tiering was explicitly stated in the source notes; blank cells indicate that tiering was not explicitly agreed for that field.

Commented [1]: Participants to add the test of the field tiers and then update this line.

Note: The Picklist Values / Notes column in Table 2 was populated during Day 3 (see Section 6.1). Both sets of decisions are presented together here for ease of reference.

Table 2: *Case/Patient domain: field-level decisions (Day 1) and controlled vocabulary assignments (Day 3).*

Field	Tiering	Discussion / Decision (Day 1)	Picklist Values / Notes (Day 3)
Case ID / Unique Patient ID		Case ID can only be assigned once a positive result is obtained. A unique patient ID will be added separately, as a patient	Unique Patient ID: Free text — to be generated by data collection tool

Field	Tiering	Discussion / Decision (Day 1)	Picklist Values / Notes (Day 3)
		can have multiple cases at different time points. Unique Patient ID is optional.	
Date of Admission	**Mandatory**	Mandatory. Date of admission to the facility where the sample was taken. Community sampling noted as a consideration — recommended in those contexts.	
Date of Onset of Symptoms	**Optional**	Optional.	
Sex / Gender		Consensus to retain biological sex only (male/female/unknown), noting sensitivities around funding constraints.	Male, Female, Unknown
Date of Birth / Age		Date of birth raises privacy concerns for data sharing. Age category to be used instead.	Age Group: 0–28 days; 1–11 months; 1–4 years; 5–9 years; 10–14 years; 15–19 years; 20–24 years; 25–59 years; 60 and above
Population Group		To be added: neonatal, paediatric, adult. To be aligned with WHO population group terms.	Neonate, Infant, Paediatric, Adult
Clinical Signs and Symptoms	**Mandatory**	Renamed from 'clinical syndrome'. Mandatory. Captures why the test (e.g. CSF) was requested.	Fever >38°C, vomiting, headache, neck stiffness, bulging fontanelle (infants), convulsions/seizures, photophobia, petechial rash, other
Vaccination Status		To be made pathogen-specific where relevant. Dropdown specific to meningitis pathogens.	Yes (with sub-picklist of vaccine types), No, Unknown Vaccine Types: MenACWY, MenAC, MenACW, ConjugateA, PCV13, Hib, Ppv23, PCV10 (GSK), PCV10 (Pneumosil), Men5/MenACWYX, 4CMenB, MenB (fHbp), Other Source of Vaccination History: Self-reported (verbal), Vaccination card, Register/medical record
Final Hospital Outcome		Discharged, fatality, unknown. Date of outcome to be added at time of recording. Defined as final	Discharged, Deceased, Still Admitted, Transferred, Absconded, Other

Field	Tiering	Discussion / Decision (Day 1)	Picklist Values / Notes (Day 3)
		outcome at end of hospitalisation.	
Sequelae at Discharge		Neurological symptoms. WHO recommends determining at end of hospitalisation. To be captured as picklist at discharge.	Hearing loss, visual loss, neurological impairment, new onset seizures, amputation and skin scarring, cognitive impairment, other
Follow-up Status	**Optional**	Optional.	Same picklist as Final Hospital Outcome. Follow-up generally at least two weeks post-discharge.
Place of Residence		Specific address raises privacy concerns. Area-level information acceptable and useful for genomic surveillance.	Free text — actual address or area of residence
Travel History	**Recommended**	Recommended. To include local and international travel.	Travel History — Place Visited: Free text — country, city, town, or village
Hospital / Facility Name / healthcare centre		To be added.	
Administrative Level		Facility-level detail needed for patient information. Administrative levels 0–3 to be collected.	
Type of Facility			Private, Public, Other (e.g. military)
Level of Care			Primary, Secondary, Tertiary, Quaternary (academic)
Case Classification			Probable, Suspected, Confirmed
Pre-Antibiotic Given			Yes, No
Was Sample Collected			Yes, No (with free text reason if No)
Province / State / County			Picklist despite initial difficulty in populating for all countries. Free text avoided to prevent inconsistency.

Commented [2]: The notes on this was not clear. Someone just wrote "Add name of hospital/healthcare center" under Day 1. So, I'm not sure if it was meant to be a data field or part of a description/definition of an existing data field

Commented [3]: I think inclusion of detailed facility information as a data field. Facility type (hospital, clinic, laboratory, health center)

Commented [4]: Agree with Rekha

Field	Tiering	Discussion / Decision (Day 1)	Picklist Values / Notes (Day 3)
City / Town / Village			Picklist
Specimen Type			CSF, Blood, Brain Swab. Brain tissue applicable to encephalitis only.
Purpose of Sampling			Diagnosis, Surveillance, Outbreak, Research
Specimen Collection Tube			Trans-isolate, Dry tube
Organism Name			*N. meningitidis, H. influenzae, E. coli, K. pneumoniae, S. pneumoniae, Listeria monocytogenes,* GBS*, Cryptococcus neoformans*
Date Format			DD/MM/YYYY calendar
Specimen ID			Free text — to link to patient without identifying them. Countries to use their own existing ID systems.
Lab Name			Free text
Contact Information			Free text — for follow-up purposes; can be hidden to protect patient privacy
Comorbidities / Risk Factors			HIV/immunodeficiency/immunocompromised, crowded living conditions, functional asplenia, complement deficiency, malnutrition, head injury/CSF leak, neonatal/congenital malformation, diabetes/metabolic syndromes, chronic liver or kidney disease, pregnancy, other

4.3 Dictionary Finalisation — Specimen Domain

The Specimen domain was reviewed during the latter part of Day 1 and continued into Day 2. The Day column indicates when each field was discussed. Where the same field was discussed across both days, both entries are included in the same row.

Note: Day 2 decisions for this domain are included in Table 3 below for continuity. They are also referenced in Section 5.1.

Table 3: *Specimen domain: field-level decisions from Days 1 and 2.*

Day	Field	Discussion / Decision
1	Specimen ID vs Sample ID	Distinction between specimen and sample ID discussed. Consensus to use specimen ID.

Day	Field	Discussion / Decision
1	Episode Number	Linked to patient/specimen but not necessarily to a positive case. The term 'episode' was noted as not well understood.
1	Volume Collected	Removed. Only important for biobanking, not for data sharing or epidemiological surveillance.
1	Sample Collected By	Removed. Always tied to the hospital, as doctors extract CSF/blood samples.
1–2	Collection Device / Specimen Collection Tube	Day 1: Renamed to 'specimen collection tube'. Dry tube for PCR and trans-isolate (TI) for culture. Collection device influences downstream analysis. Day 2: Consensus reached on this name. Dry tube for PCR; TI for culture. WHO recommends TI if transport time exceeds four hours.
1	Collection Method	Not considered necessary as a separate field; emphasis placed on the specimen itself.
1	Enrichment Type	Removed due to urgency of sample processing requirements.
1	Specimen Condition	To be included.
1	Storage Temperature	Optional. Considered good lab practice but not essential for data sharing.
1	Date Received by Lab	Important due to transport time. Time to be added as specimens should be processed within one hour of receipt.
1–2	GPS / Geolocation	Day 1: Coordinates not to be added to the specimen domain. Geolocation of patient's place of residence to be placed in the patient section. Day 2: Contested in terms of practicality. Added as optional. Can be calculated from hospital coordinates. Noted as useful for epidemiological heatmapping and clustering. Linked to residential address flagged as a privacy concern.
1	Referral Lab Name	To be added as mandatory.
1	Environmental Sampling	Not considered in scope.
1	Organism / Isolate	To be included. Clinical variables to be separated into their own domain.
2	Pre-Antibiotic Given	Recommended.
2	Date of Specimen Collection	Renamed from 'date of sample collection'.
2	Personal Identifiers	Suggested for internal use only.
2	Lab Name	Mandatory. Includes testing lab only at the diagnostic testing stage. Referral lab name also included.

Commented [5]: Please check: One notetaker said "coordinates will not be added." Another notetaker said "consensus was not fully reached; however, it was added as optional." The third one said "added coordinates (latitude and longitude) as optional"

Commented [6]: I think the consensus was to add this as "Optional". Where possible to add the Geolocation coordinates of the patient and specimen which provides important epidemiological data in the downstream analysis.

5. Day 2 — 05 May 2026

5.1 Specimen Domain — Continued

Day 2 began with a recap of Day 1 before continuing the review of the Specimen domain. The Day 2 decisions for the Specimen domain are incorporated into Table 3 (Section 4.3) for continuity.

5.2 Diagnostic Tests Domain

Commented [7]: There were no notes on tiering

The group then worked through the Laboratory / Diagnostics domain in detail. The agreed workflow follows the order in which tests are performed in the laboratory: macroscopy and microscopy, followed by biochemistry, antigen testing, culture, organism identification, susceptibility testing, serotyping, and then molecular and sequencing methods. Key decisions are summarised in Table 4 below.

Table 4: *Laboratory / Diagnostics domain: agreed fields and decisions from Day 2.*

Category	Fields and Decisions
Macroscopic Results	Visual appearance of specimen. Recommended. Agreed that macroscopy enhances diagnosis even if not determinative.
Microscopic Results (Cytology)	Cell count, Gram stain (GNB, GPC, GNC, GPB), India ink, polymorphs, erythrocytes, lymphocytes, neutrophils, RBCs, WBCs, yeast cells.
Biochemical Tests	CSF glucose, serum glucose, CSF protein, CSF lactate, CSF chloride.
Antigen Tests	CrAg (+/-), Bacterial Pastorex — NmA, NmC, NmY/W135, GBS, E. coli K1. Mandatory even given lateral flow test unreliability; Latex test (bacterial) important post-microscopy.
Microbiological Culture	Positive/negative/not done. Applicable to bacteria and fungi. Organism ID follows culture.
Organism Identification Method	Manual biochemical, automated biochemical, mass spectrometry, molecular.
AST Method	Moved before AST results to reflect logical lab workflow. Disk diffusion, broth microdilution, gradient diffusion (E-test). Confirmed as phenotypic AST methods.
AST Standards	EUCAST, CLSI, CASFM. CASFM followed in West Africa. Added together but distinguished.
AST Results	Comprehensive drug list per organism. MIC values to be free text (µg/mL). Resistance phenotype included with organism-specific phenotypes.
Serotyping / Serogrouping	Per pathogen/organism: *N. meningitidis* (NmA–NmX), *H. influenzae* (HiB, HiNT, HiNb), *E. coli* (K1), *S. pneumoniae* (vaccine and clinically relevant serotypes), GBS. Free text option for unlisted serotypes.
Cefinase Test	Added as a new variable.
Resistance Phenotype	Pathogen-specific. Includes Penicillin-Non-Susceptible *Pneumococci*, MLSB, Third-Generation Cephalosporin Resistance (ESBL/AmpC), CRE, BLPAR, BLNAR.

Category	Fields and Decisions
qPCR / Molecular Testing	qPCR result (presence/absence) and CT values (free text). Applicable to bacterial and viral pathogens. Real-Time PCR and End-Point PCR methods included. qPCR typically conducted at reference labs.
Sequencing Fields	Nucleic acid extraction method (recommended), QC quantification and integrity/quality (mandatory), sequencing lab name, sequencing date, sequencing approach, library preparation kit, sequencing platform, read type, R1 fastq filename (recommended), BAM files (optional), VCF/BED files.
Viral Pathogen Names	Noted as a to-do action for inclusion.

5.3 Bioinformatics Domain

The Bioinformatics domain was reviewed and field tiering was agreed during Day 2. The Tiering column is populated only for fields where tiering was explicitly agreed during this session. Fields that appear only in the Day 4 sequencing picklist table have a blank tiering cell.

Note: The Format / Picklist Values column in Table 5 was populated during Day 4 (see Section 7.1). Both sets of decisions are presented together here for ease of reference.

Table 5: *Bioinformatics domain: field tiering (Day 2) and format/picklist values (Day 4).*

Field	Tiering	Notes (Day 2)	Format / Picklist Values (Day 4)
Total reads	Recommended		Numeric
Depth of coverage	Mandatory	Can be calculated from de novo assembly	Sequencing depth of coverage: Numeric
Breadth of coverage	Mandatory		Sequencing breadth of coverage: Numeric
N50	Recommended		Numeric
L50	Recommended		Numeric
GC content	Recommended		Numeric
Mapped reads	Recommended		Numeric
Number of contigs	Recommended		Numeric
Dehosting method	Mandatory		Free text
Assembly software name	Mandatory		Assembly software (de novo): SPAdes, Velvet, Unicycler, Flye, Skesa, Canu Assembly software (reference-based): MiniMap, BWA, Bowtie, Ivar

Commented [8]: Is this supposed to be "% of mapped reads"?

Commented [9]: Method or Bionformatics tool?

Commented [10]: Both method__(mentioning kit used) and bioinformatic tool__mentioning what tool was used

Field	Tiering	Notes (Day 2)	Format / Picklist Values (Day 4)
Assembly software version	Mandatory		Numeric
Consensus sequence software and version	Optional		Free text
Bioinformatics protocol	Free text summary		Free text
MLST	Recommended	Bacteria and fungi	MLST type: Alphanumeric
cgMLST	Recommended	Pathogen-specific	cgMLST assignment: Alphanumeric
Bioinformatics serotyping	Recommended	GBS, S. pneumoniae, GAS	
Virulence genes	Mandatory	Pathogen-specific picklist	
AMR genes	Mandatory	Pathogen-specific; may include plasmid-borne genes	
AMR prediction database name	Mandatory		
AMR genes detected — database and version	Mandatory		
Vaccine targets	Recommended		
GPSC classification	Recommended	S. pneumoniae	
Clonal complexes	Recommended	Candida assigned clades for phylogeny	
Nucleic acid extraction method			Free text (with picklist to be added)
QC — quantification method			Free text
QC — integrity/quality method			Free text
Sequencing lab name			Free text
Sequencing date			Numeric/date
Sequencing approach			WGS; Metagenomics; Targeted (Hybrid capture; Amplicon-based: 16S, 18S, ITS); Sanger sequencing
Library preparation kit			Recommended. Illumina Nextera, NEB Next, ONT Rapid Barcoding Kit, ONT

Commented [11]: Would this be an issue if they only submit the consensus sequence to a repository?

Commented [12]: I dont think that would be an issue- thus "Optional"

Commented [13]: This is very high-level. Does this mean the typing result or tool?

Commented [14]: I assume this will be linked to an actual calendar

Commented [15]: Yes

Field	Tiering	Notes (Day 2)	Format / Picklist Values (Day 4)
			Ligation Sequencing Kit, Illumina DNA Flex, Other (free text)
Sequencing target depth			Free text numeric. Changed to recommended as 100× cannot always be achieved.
Read type			Single-end, Paired-end
Read length			Long read, Short read
R1 / R2 fastq filename			Free text
BAM file name			Free text
VCF file name			Free text
Assembly method			De novo, Reference-based

Commented [16]: Can this be confused by actual values of the gene length? May be better to just stick to the "sequencing technology type", such as long-read, short read, sanger sequencing, or "sequencing platform type" such as Illumina, Nanopore, etc, which were already captured elsewhere.

5.4 Additional Decisions

The following additional decisions were made during Day 2:

- The scope of the standard is confirmed as community-acquired.
- Case classification was confirmed as needed under the Surveillance Context domain.
- Data governance: the group agreed that sharing agreements should be established before metadata is collected.
- BioProject accession: recommended. BioSample accession: recommended. AGARI accession: recommended. GenBank accession: included.

Commented [17]: Not sure under which domain this was meant to go

6. Day 3 — 06 May 2026

6.1 Joint Session with Neonatal Sepsis Group and Controlled Vocabulary Assignment

The day began with a joint session between the meningitis and neonatal sepsis working groups to consolidate progress and identify variables that could be shared between the two templates. Picklists from the sepsis template were borrowed as a starting point for the meningitis template where applicable.

The group then worked through the Case/Patient domain to assign controlled vocabularies and picklists to all applicable fields. These picklist values are incorporated into the Picklist Values / Notes column of Table 2 (Section 4.2) for ease of reference.

6.2 Additional Decisions from Day 3

The following additional decisions were also reached during Day 3:

- A separate tab in the reporting form to be created for identifiable patient information (e.g. patient name, date of birth), linked to the main dataset. This tab will not be shared when data is distributed.
- Guidance text to be added to the travel history field.
- Vaccination status (yes/no/unknown) to be made mandatory.
- Final hospital outcome option to be changed from 'discharged' to 'discharged home'.
- AST data management was revised: each isolate remains in a row, with data organised into blocks by common factors such as testing date and method. Each column represents a different antimicrobial result.
- The redundant qPCR method field (in-house/commercial) was removed from the template.
- Urban/rural to be added as a new variable for surveillance purposes.

Commented [18]: The notes were not clear under which domains these data fields were meant to be edited or how they should be arranged -especially the AST data management

Commented [19]: Vaccination status to go under vaccination history and that each region will have specific vaccines to that region as a picklist

Commented [20]: AST data will go under AST results Different organisms have different antibiogram, so each pathogen will be treated accordingly based on its antibiogram and antibiotics will be selected from a picklist. However, the common entires such as testing date/method will remain constant ad at the top

7. Day 4 — 07 May 2026

7.1 Sequencing and Bioinformatics Picklists

Day 4 continued the population of picklists for the sequencing and bioinformatics domains. The group noted a target of completing the metadata template by end of May 2026, with a first draft of the manuscript by end of July 2026.

The sequencing field picklists are incorporated into the Format / Picklist Values column of Table 5 (Section 5.3) for ease of reference. The sequencing platform picklist is noted separately below.

A comprehensive list of sequencing instruments was populated to accommodate retrospective data and future platforms. Instruments were organised by manufacturer. An 'Other (free text)' option was included for each manufacturer category to accommodate less common platforms. Manufacturers included: Illumina, Pacific Biosciences, Ion Torrent, Oxford Nanopore, BGI/MGI Genomics, Element BioSeq, and Sanger sequencing instruments.

7.2 Bioinformatics Serotyping — Gene Lists by Organism

The bioinformatics serotyping picklists were reviewed and finalised. The group prioritised genes conferring clinically relevant serotypes; housekeeping genes used for species identification were excluded. Genes were grouped under their respective serogroups. Where the list was too extensive, an 'Other (free text)' option was included. Key decisions on serotyping gene lists are summarised in Table 6 below.

Table 6: *Bioinformatics serotyping gene lists by organism (Day 4).*

Organism	Serogroups / Serotypes	Key Serotyping Genes
Neisseria meningitidis	NmA, NmB, NmC, NmW/NmW135, NmX, NmY	ctrA (species ID); NmA: csaB, sacB; NmB: csb, synD; NmC: csc, synE; NmW: csw, synG; NmX: csxB, xcbB; NmY: csy, synF
Haemophilus influenzae	Hib, HiNb (a, c, d, e, f), HiNT (bexA negative)	Hib: capb (bcsA, bcsB, bcsC); HiNb-a: capa (siaA–D); HiNT: no capsule gene
Escherichia coli	K1 only (most associated with meningitis). Other O-antigen serotypes not considered for this exercise.	neuC, neuD, neuB, neuE, neuS, neuA, kpsMTII, kpsII, mprA, lrhA, rfaH — subject to further review
Streptococcus pneumoniae	Vaccine and clinically relevant serotypes listed. 'Other' option with free text included.	cps_locus genes per serotype (e.g. wzg for serotype 1, wzh for serotype 3)
GBS	Ia, Ib, II, III, IV, V, VI, VII, VIII, IX, X, NT	cps genes per serogroup (e.g. Ia: cpsH; III: cpsG)
Cryptococcus neoformans	Molecular types A (VNI, VNII, VNB), D (VNIV), AD (VNIII), B&C (VGI–VGIV)	
Listeria monocytogenes	1/2a, 1/2b, 1/2c, 4b, Other	Based on LisSero bioinformatics tool. Added as a pathogen as it is predominantly associated with neonatal meningitis.

7.3 Virulence Genes

Virulence gene lists were reviewed per pathogen. The *E. coli* virulence gene section was left incomplete due to the extensiveness of the list and will be completed at a follow-up meeting. The following were documented during the session:

- *N. meningitidis*: porA, fetA, porB, fHbp, nhba, nadA
- *H. influenzae*: lipopolysaccharide genes (lic1, lic2A, lic3, lgtF, lpsA, lex2); adhesion proteins (hap, hia, hsf); pili (hifA, hifE); protein E (pe); IgA protease (igaA, igaB); biofilm formation (pilA, wecA, siaA, siaB); phase variation (galE)
- *E. coli*: left for a later meeting due to the extensiveness of virulence factors that could not be covered in the time available.

7.4 AMR Gene Lists by Organism

AMR gene lists were reviewed and agreed per pathogen. Genes are organised by drug class. The group noted that aminoglycosides and macrolides do not readily cross the blood-brain barrier and are therefore considered optional for meningitis. The agreed AMR gene lists are summarised in Table 7 below.

Table 7: *AMR gene lists by organism and drug class (Day 4).*

Organism	Drug Class	Key AMR Genes
N. meningitidis	Beta-lactams	penA, ponA, porB, blaROB-1
	Cephalosporins	penA795, penA327
	Fluoroquinolones	gyrA, gyrB, parC mutations
	Rifamycin	rpoB mutations
	Macrolides	mtrR mutations
	Tetracyclines	rpsJ
	Sulphonamides	folP
	Efflux pump	mtrR mutations
H. influenzae	Beta-lactams	blaTEM-1B, blaROB-1, ftsI, ompP2
	Fluoroquinolones	gyrA, parC
	Macrolides	mefA, mefE, ermA, ermB, msr(A/B/D), L4, L22
	Rifamycin	rpoB
	Sulphonamides	folA, folP
	Efflux pump	acrA, acrB, acrR, tolC
E. coli	Beta-lactamases	blaCTX-M, blaTEM-1, blaTEM-1B, ampC, blaNDM-5, blaVIM, blaIMP, blaKPC, blaOXA-48
	Fluoroquinolones	qnrA, qnrB, qnrS, gyrA, parC
	Aminoglycosides	aac(3)IIa, armA, rmtB, ant, aph(3), strA, strB
	Tetracyclines	tetA, tetB
	Colistin	mcr1, mcr10
	Sulphonamides	sul1, sul2, dfrA
S. pneumoniae	Beta-lactams	pbp1a, pbp2x, pbp2b
	Macrolides & lincosamides	ermB, mefA/E, msrD
	Tetracyclines	tetM, tetO
	Fluoroquinolones	gyrA, parC, parE
	Sulphonamides	folP
Listeria monocytogenes	Tetracyclines	tet(M), tet(S), tet(L)
	Aminoglycosides	aacA-aphD
	Glycopeptides	mecC

Organism	Drug Class	Key AMR Genes
	Chloramphenicol	cat
	Beta-lactams	pbp, ampC
	Sulphonamides	sul1, sul2
Cryptococcus neoformans	Azoles	ERG11
	Azole efflux	AFR1
	5-Fluorocytosine resistance	FUR1, FCY1, FCY2
	Amphotericin B	ERG11, ERG2, HOB1
	Echinocandin	fks1, cdc50
GBS	Beta-lactams	pbp1a, pbp2x, pbp2b
	Macrolides & lincosamides	ermB, ermA, ermT/R, mefA/E, lsa, msrD, mefA-msrD, aac(6')-aph(2")
	Tetracyclines	tetM, tetO, tetL, tetT, tetW
	Fluoroquinolones	gyrA, parC

8. Closing Remarks and Next Steps

- Stakeholders to review and highlight which AMR and virulence genes should be mandatory, recommended, or optional.
- Stanford Kwenda to add field descriptions. Metadata template to be completed by end of May 2026.
- First draft of manuscript targeted for end of July 2026, with publication targeted for end of July 2026.
- Folder to be shared with all participants for manuscript contributions.
- Ontology mapping to be completed, followed by a short review meeting.
- Follow-up meetings to be held virtually (Zoom or Teams) to review the template and finalise content for manuscript publication.